

Abdulwahab Felemban

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PhD candidate (expected 2027) specializing in multimodal LLMs, VLMs, and VLAs for autonomous driving, motion prediction, and safety-aware planning.

RESEARCH INTERESTS

- Multimodal LLMs, VLMs, and VLAs for autonomous driving perception, planning, and generation
- Motion prediction, trajectory generation, world models, and structured reasoning in dynamic scenes
- Safety-aware instruction tuning and evaluation for autonomy systems
- Large-scale autonomous-driving data curation, benchmarking, and closed-loop analysis

PUBLICATIONS

- **Felemban, A.**, Hroub, N., Ding, J., Fu, Y., Hammoud, H. A. A. K., Bezerra, L. C. D., Khan, F. F., Zhang, W., Elhoseiny, M. *DriveGavel: A Zero-Shot VLM Safety Layer for Autonomous Driving Planners* [Under review, AAAI 2027, advanced to Phase 2. Manuscript available upon request]
- **Felemban, A.**, Hroub, N., Ding, J., Zhang, W., Khan, F. F., Elhoseiny, M. *Wider Coverage, Better Plan: Admissible Alternatives for End-to-End VLA Planning* [Under review, ICLR 2027. Manuscript available upon request]
- **Felemban, A.**, Hroub, N., Ding, J., Bakr, E.M., Shen, X., Mohamed, A., Elhoseiny, M. *iMotion-LLM: Motion Prediction Instruction Tuning* [WACV 2026 [🔗](#)]
- Gou, C., **Felemban, A.**, Khan, F.F., Zhu, D., Cai, J., Rezatofighi, H., Elhoseiny, M. *How Well Can Vision Language Models See Image Details?* [ArXiv:2408.03940, 2024 [🔗](#)]
- **Felemban, A.***, Battach, Y.* (Equal contribution), Khan, F. F., Radwan, Y. A., Li, X., Marchese, F., Beery, S., Jones, B. H., Benzoni, F., Elhoseiny, M. *ReefNet: A Large-Scale Dataset and Benchmark for Fine-Grained Coral Reef Recognition* [NeurIPS 2026, Evaluations & Datasets Track, [arXiv:2510.16822](#) [🔗](#)]
- Khan, F. F., Radwan, Y., Bakr, E.M., **Felemban, A.**, Mir, A., Michiels, N. K., Temple, A. J., Berumen, M. L., Elhoseiny, M. *FishNet 2.0: A Comprehensive Multimodal Benchmark for Long-Tailed Fine-Grained Fish Understanding* [Under review, ICLR 2027; earlier version [arXiv:2509.25564](#) [🔗](#)]
- Al-Naffouri, T.Y., Masood, M., **Felemban, A.**, Aljoud, M.F., Bader, A., Alkhodary, M.T., Ahmed, T.B.K. *Real-Time Control of Mechanical Valves Using Computer Vision and Machine Learning* [US Patent App. 18/021,241, 2023 [🔗](#)]
- **Felemban, A.** *Deep Learning Action Anticipation for Real-Time Control of Water Valves: Wudu Use Case* [MS Thesis, 2021]
- AlSharif, M.H., Ahmed, M., **Felemban, A.**, Zayat, A., Muqabel, A., Masood, M., Al-Naffouri, T.Y. *Robust 2D Indoor Positioning Algorithm in the Presence of Non-Line-of-Sight Signals* [EUSIPCO 2020 [🔗](#)]

SELECTED PROJECTS

DriveGavel: Zero-Shot VLM Safety Layer for Driving Planners

- Training-free VLM judge wrapping any planner: reads deterministic collision evidence, approves or overrides with a pre-verified evasive maneuver; collisions 83.3% → 13.3% on NeuroNCAP closed-loop.

Wider Coverage, Better Plan (formerly iMotion-VLM): VLA Planning with Admissible Alternatives

- Trained a VLA planner's head on admissible alternative plans accepted by a rule-based scorer, plus a reinforcement-learning stage checked against a speed-matched baseline: NAVSIM v2 navhard single-plan EPDMS 0.315 → 0.373, best-of-16 0.359 → 0.614.

iMotion-LLM, InstructNuPlan / InstructWaymo Datasets

- Early work (begun late 2023, preprint June 2024) coupling an LLM with trajectory prediction: safety- and feasibility-aligned instruction datasets over vectorized BEV representations, with instruction adherence evaluated end to end.

ReefNet (925K Annotations): Large-Scale Dataset and Benchmark

- 925K-annotation coral benchmark built with marine ecologists; ViT/BEiT/BioCLIP/CLIP/SigLIP baselines and a verification pipeline.

Smart-Tap: Fully Integrated AI System for Water Valve Control

- Real-time action-anticipation prototype with pose estimation and spatio-temporal GCNs.

EDUCATION

PhD	King Abdullah University of Science and Technology (KAUST) , Electrical & Computer Engineering, Saudi Arabia	01/2022 - present
	• GPA: 3.79/4.0 • 06/2023 - present: Prof. Mohamed Elhoseiny. Vision-language-action planning and closed-loop safety evaluation for autonomous driving, and large-scale benchmarks (ReefNet) • 01/2022 - 05/2023: Prof. Tareq Al-Naffouri. Human action anticipation for real-time control	
MS	King Abdullah University of Science and Technology (KAUST) , Electrical & Computer Engineering, Saudi Arabia	01/2020 - 01/2022
	• GPA: 3.83/4.0 • Advisor: Prof. Tareq Al-Naffouri • Focus: Deep Learning and Human Action Prediction	
BS	King Fahd University of Petroleum and Minerals (KFUPM) , Electrical Engineering, Dhahran, Saudi Arabia	08/2014 - 12/2019
	• GPA: 3.43/4.0 • Concentration: Signal Processing and Wireless Communication • Senior Project: Real-Time AI Surveillance System	
-	Georgia Institute of Technology , Exchange Student, Atlanta, GA, USA	08/2018 - 12/2018

SKILLS

Multimodal / VLM / VLA: Qwen-VL, LLaVA, CLIP/SigLIP, instruction tuning (SFT/DPO/GRPO), multimodal reasoning.
Autonomous Driving / Motion: BEV and trajectory representations, perception-prediction-planning pipelines, open/closed-loop evaluation, Waymo/nuScenes/nuPlan workflows.
Training & Systems: PyTorch, multi-GPU training, Slurm, vLLM inference, large-scale preprocessing.
Programming: Python, OpenCV, MATLAB, R, TensorFlow (basic), C++ (basic).
Deep Learning: Vision transformers, temporal and generative models, end-to-end AD systems, graph deep learning.
Additional Skills: Data curation, metadata cleaning, parallel processing, digital signal/image processing.

OTHER ACADEMIC CONTRIBUTIONS

ELLIS Summer School on Autonomous Driving (SSAD 2026) , Barcelona. Selected participant. Poster on DriveGavel and admissible-alternative VLA training. Discussions with Maximilian Igl, José M. Álvarez (NVIDIA), Holger Caesar (TU Delft), Kashyap Chitta, Alexandre Boulch (Valeo), Puneet Dokania (Five AI).	Sep. 15–18, 2026
Selected Student Presenter: KAUST Rising Stars in AI Symposium 2026. Organizers: Jürgen Schmidhuber and Bernard Ghanem.	Feb. 9–11, 2026
Reviewer: WACV 2026, ICCV2025 Workshop Proceedings, ICCV 2025 Workshop CV4E, ACML 2024, SIGGRAPH Asia 2024.	
Workshop Organizer: CVPR 2024 – C3DV: 2nd Workshop on Compositional 3D Vision.	06/2024

AWARDS & FUNDING

CEMSE Dean's List Award 2021/22 (top 20% of students)
\$1M in Research Funding: NEOM Ocean Science and Solutions (OSSARI) Research Fund (\$600k, 2024); KAUST Translational Fund (\$450k, 2022).
Projects Awards: Start Smart [🔗](#) (1st place, \$9k, EcoStream, 2024), Digital Innovation Award [🔗](#) by the Saudi Ministry of Communications and IT (1st place, Smart-Tap, \$21k, 2022), KFUPM Undergraduate Senior-Project Award (1st place, AI Surveillance Project, 2019), SSI Internship Poster Competition (2nd place, Indoor localization project, 2019).

EXPERIENCE

KAUST Academy 1-Week AI Specialization Instructor: Taught undergraduates the fundamentals of AI (ML, CV, neural networks, Python).	01/2025
KAUST Saudi Summer Internship (SSI): Ultrasonic indoor localization, filtering non-line-of-sight signals with machine learning.	05/2019 - 08/2019
KFUPM IEEE Student Branch Publicity Chair: events and seminars.	01/2018 - 01/2019